

CSCI 361 Lecture 14:

Undecidability Reductions

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Announcements & Logistics

- **HW 7** due tomorrow night
 - No extensions on this HW!
 - Solutions will be released by Thursday
- Reminder **Midterm 2**: Thursday, Nov 6
 - Topics in Chapter 2 onwards (HW 4-7) will be the focus
 - Similar format as Midterm 1
 - Will release practice problems
- **New office hours**: Fri 1-2.30 pm, Mon 2-3.30 pm, Wed 10.30 am-noon and by appointment
 - Feel free to stop by office at other times or email me for a meeting

Story So Far

- We're finding the boundary between decidability and undecidability
 - Building intuition on when problems are solvable
- Focus on positive and negative reductions:
 - **Positive reduction**: when you prove a language A is decidable using a decider for another decidable language B as a subroutine
 - **Negative reduction**: when you prove a language B is undecidable by reducing from a known undecidable problem A

Last Time

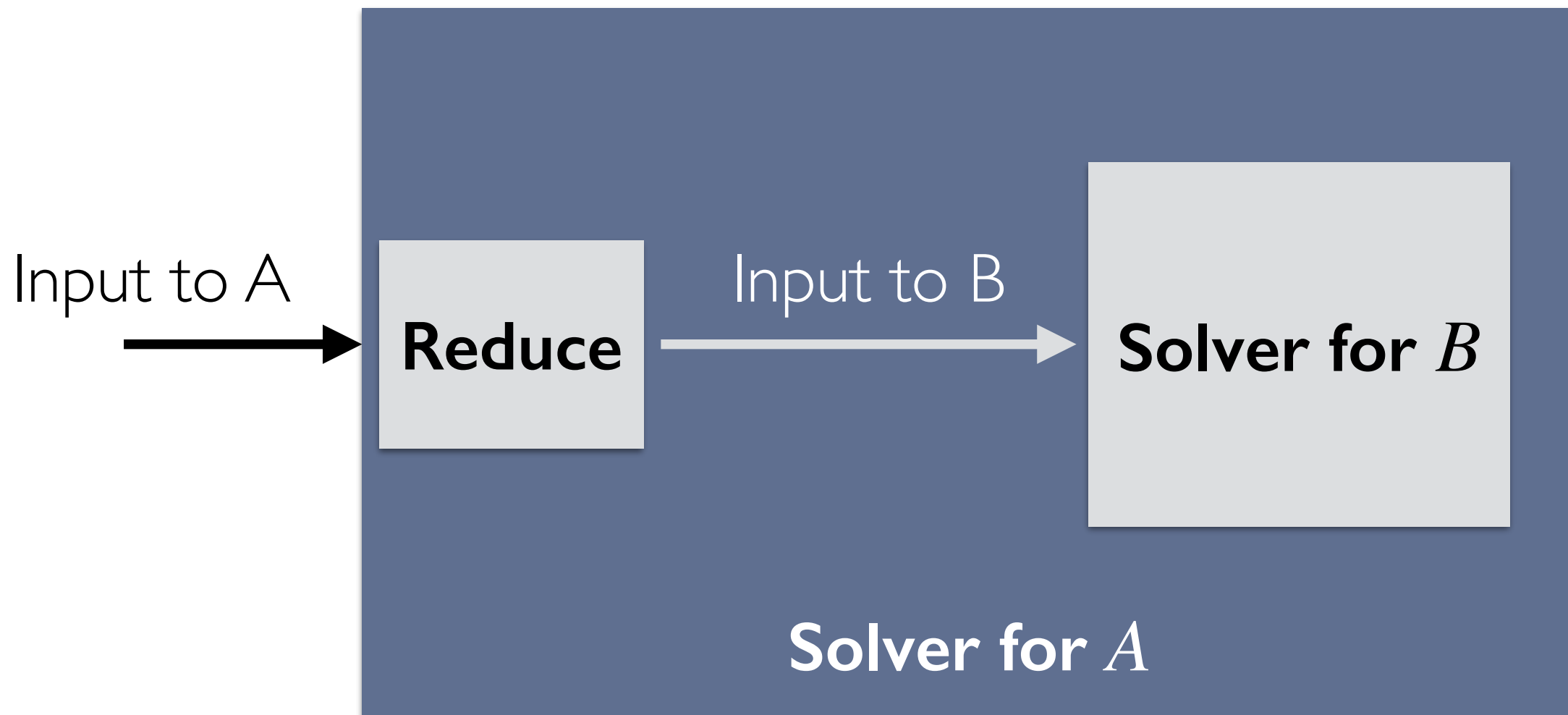
- Proved some problems are undecidable:
 - Given a TM and an input, does it accept it?
 - $A_{TM} = \{ \langle M, w \rangle \mid M \text{ is a TM and } w \in L(M) \}$
 - Given a TM and an input, does it halt on it (accept/reject it)?
 - $HALT_{TM} = \{ \langle M, w \rangle \mid M \text{ is a TM and } M \text{ halts on } w \}$
 - Given a TM, is its language empty?
 - $E_{TM} = \{ \langle M \rangle \mid M \text{ is a TM and } L(M) = \emptyset \}$

Today

- Do more practice with reductions to prove undecidability
- Identify many more TM undecidable and TM unrecognizable problems

Reduction Review

- Informally, problem A **reduces** to Problem B if we can use the solution of B to solve A



Review Exercises

Problem. Show that

$EQ_{TM} = \{ \langle M, N \rangle \mid M, N \text{ are TMs and } L(M) = L(N) \}$ is undecidable.

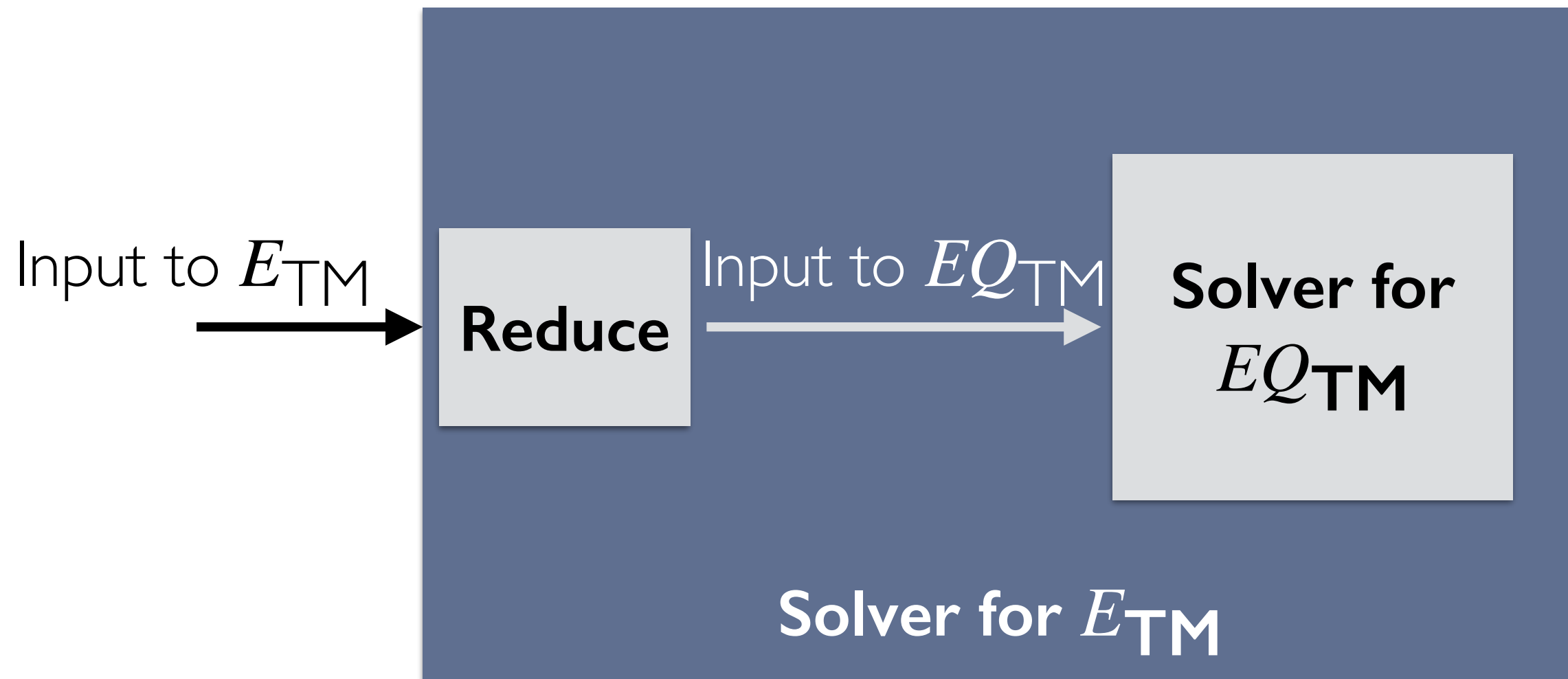
Hint. Reduce E_{TM} to it.

Problem. Show that

$FINITE_{TM} = \{ \langle M \rangle \mid M \text{ is a TM and } L(M) \text{ is finite} \}$ is undecidable.

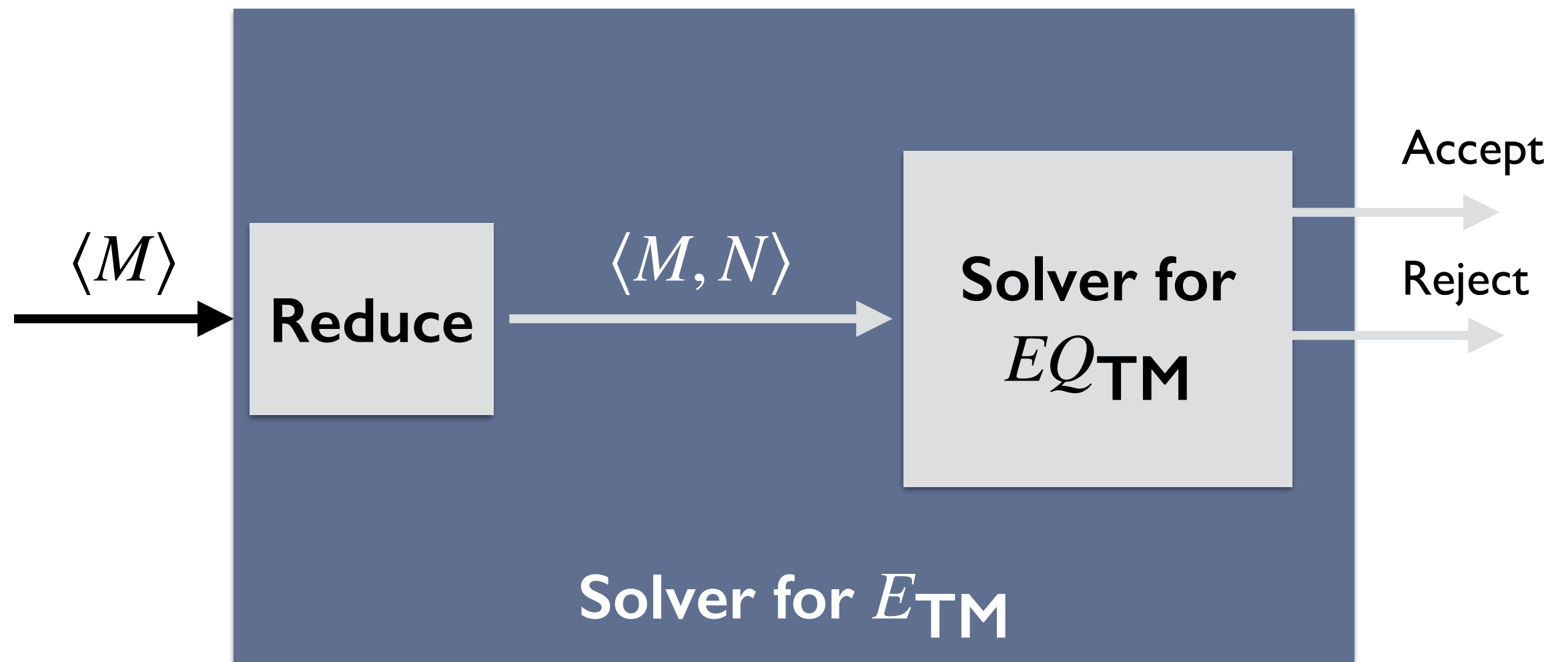
EQ_{TM} is undecidable

Reduction from E_{TM}



EQ_{TM} is undecidable

Reduction step: Keep $\langle M \rangle$ as is, create TM N that rejects all inputs and thus $L(N) = \emptyset$



Correctness: $L(M) = \emptyset$ if and only if $L(M) = L(N)$

FINITE_{TM} is undecidable

Reduce from A_{TM} :

- Given input $\langle M, w \rangle$, create a new TM M_w such that M accepts w if and only if $L(M_w)$ is finite
- Let M_w be defined as follows:
 $M_w =$ "On input x ,
 - Run M on w , if M accepts w , accept (input x)
 - Otherwise, reject (input x)
- M accepts w iff $L(M_w) = \Sigma^*$
- M rejects w iff $L(M_w) = \emptyset$

FINITE_{TM} is undecidable

Reduce from A_{TM} :

- Given input $\langle M, w \rangle$, create a new TM M_w such that M accepts w if and only if $L(M_w)$ is finite
- Let M_w be defined as follows:
 $M_w =$ "On input x ,
 - Run M on w , if M accepts w , accept (input x)
 - Otherwise, reject (input x)
- Run decider for FINITE_{TM} on M_w , if it accepts, then reject; if it rejects, then accept

More Practice with Reductions

Definition. $\text{REGULAR}_{\text{TM}} = \{ \langle M \rangle \mid L(M) \text{ is regular} \}$. M is a TM.

Question. Show that $\text{REGULAR}_{\text{TM}}$ is undecidable.

Proof. Reduce A_{TM} to $\text{REGULAR}_{\text{TM}}$.

Goal: Given $\langle M, w \rangle$, convert it to a new Turing machine M_{new} s.t.

M accepts w if and only if $L(M_{\text{new}})$ is regular.

Idea: Let's try a similar idea as the last reduction

REGULAR_{TM} is undecidable

Proof. Let R be a decider for REGULAR_{TM}. Then consider TM D :

$D =$ "On input $\langle M, w \rangle$

1. Create $M_{\text{new}} =$ "On input x ,

1. If x has the form $0^n 1^n$, then accept.

2. Otherwise, run M on w and accept if M accepts."

2. Run R on M_{new} .

3. If R accepts, accept. If R rejects, reject.

- What is $L(M_{\text{new}})$?

REGULAR_{TM} is undecidable

Proof. Let R be a decider for REGULAR_{TM}. Then consider TM D :

$D =$ "On input $\langle M, w \rangle$

1. Create $M_{\text{new}} =$ "On input x ,
 1. If x has the form $0^n 1^n$, then accept.
 2. Otherwise, run M on w and accept if M accepts."
 2. Run R on M_{new} .
 3. If R accepts, accept. If R rejects, reject.
- Suppose M accepts w , then M_{new} accepts all x , $L(M_{\text{new}}) = \Sigma^*$
 - Suppose M does not accept w , then M_{new} only accepts $0^n 1^n$

REGULAR_{TM} is undecidable

Proof. Let R be a decider for REGULAR_{TM}. Then consider TM D :

$D =$ "On input $\langle M, w \rangle$

1. Create $M_{\text{new}} =$ "On input x ,
 1. If x has the form $0^n 1^n$, then accept.
 2. Otherwise, run M on w and accept if M accepts."
 2. Run R on M_{new} .
 3. If R accepts, accept. If R rejects, reject.
- M accepts w if and only if $L(M_{\text{new}})$ is regular. ■

Mapping Reducibility

- A technical formulation of reducibility that lets us prove more things
- **Definition.** Language A is mapping reducible to language B , denoted $A \leq_m B$, if there exists a computable function $f: \Sigma^* \rightarrow \Sigma^*$, such that

$$w \in A \iff f(w) \in B \quad \text{for every } w$$

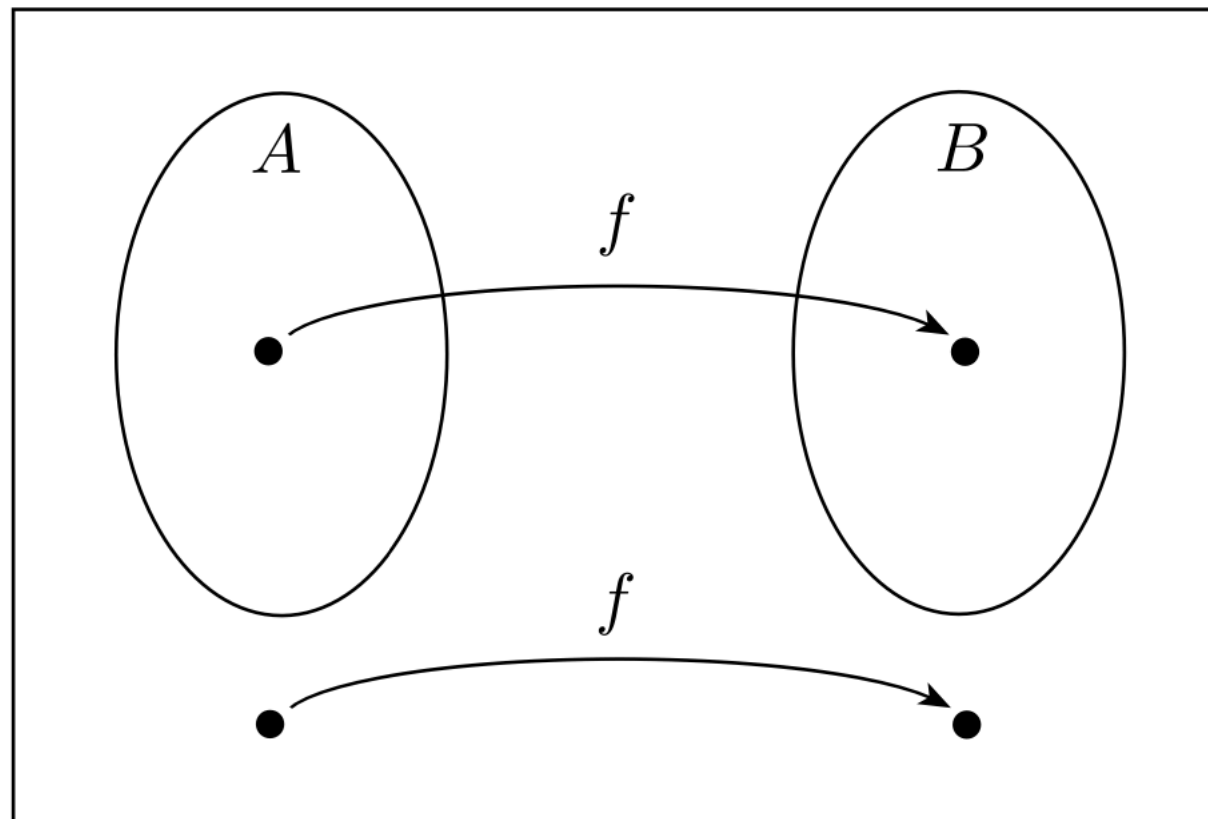
- The function f is called the **reduction** from A to B
- A function $f: \Sigma^* \rightarrow \Sigma^*$ is **computable** if some Turing machine M when given any input w , halts with just the output $f(w)$ on its tape.

Mapping Reducibility

- **Definition.** Language A is mapping reducible to language B , denoted $A \leq_m B$, if there exists a computable function $f: \Sigma^* \rightarrow \Sigma^*$, such that

$$w \in A \iff f(w) \in B \quad \text{for every } w$$

- **Remark.** If $A \leq_m B$ then $\bar{A} \leq_m \bar{B}$



Mapping Reducibility

- Using reductions to prove **decidability**:
 - **Theorem.** If $A \leq_m B$ and B is decidable, then A is decidable.
 - Why is this true?
- Using reductions to prove **undecidability**:
 - **Corollary.** If $A \leq_m B$ and A is undecidable, then B is undecidable.

Revisit Past Reductions

Reduction from A_{TM} to HALT_{TM} from last lecture:

- Suppose TM R decides HALT_{TM} .
- Construct a decider S for A_{TM} =
"On input $\langle M, w \rangle$,
 - Run R on $\langle M, w \rangle$.
 - If R rejects, then reject.
 - If R accepts, then simulate M on w . If M enters accept state, then accept; if M enters reject state, then reject.

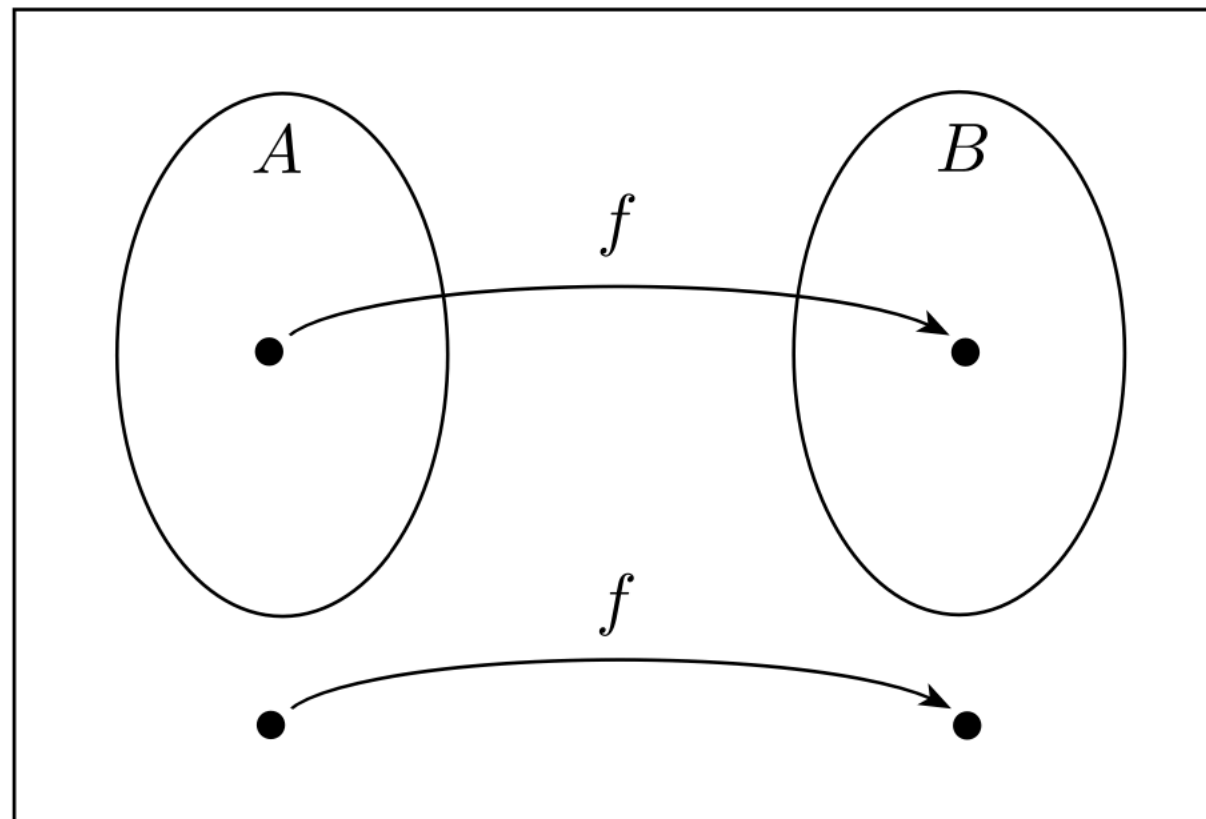
Question. Is this a mapping reduction from A_{TM} to HALT_{TM} ?

Revisit Past Reductions

Key difference: Need to map "yes" instances to "yes" and "no" to "no".

Need a computable function f that maps $\langle M, w \rangle$ to $\langle M', w' \rangle$ such that

$$\langle M, w \rangle \in A_{\text{TM}} \text{ iff } \langle M', w' \rangle \in \text{HALT}_{\text{TM}}$$



Mapping Reduction: A_{TM} to HALT_{TM}

Reduction function computed by the following Turing machine:

$F =$ "On input $\langle M, w \rangle$:

1. Construct the machine $M' =$ "On input x :
 1. Run M on x .
 2. If M accepts, *accept*.
 3. If M rejects, go into an infinite loop.
4. Output $\langle M', w \rangle$ "

Why Mapping Reductions?

- Seem unnecessarily strict, can use informal reductions just fine to prove undecidability
- Why force mapping from yes instances to yes, no to no?
 - Useful to reason about Turing recognizability and unrecognizability
- Mapping reductions to prove **recognizability**:
 - **Theorem.** If $A \leq_m B$ and B is recognizable, then A is recognizable.
- Mapping reductions to prove **unrecognizability**:
 - **Corollary.** $A \leq_m B$ and A is unrecognizable, then B is unrecognizable.

Exercise

- Review the reductions from earlier:
 - From A_{TM} to E_{TM}
 - From E_{TM} to EQ_{TM}
 - From A_{TM} to $\text{REGULAR}_{\text{TM}}$
- **Questions.**
 - Which of these are mapping reductions?
 - Is it possible to have a mapping reduction in all these cases?

Review Reduction: A_{TM} to E_{TM}

Suppose TM R decides E_{TM} . Consider the following decider:

- $D =$ "On input $\langle M, w \rangle$
 - Encode a TM M_w that does the following:
 - $M_w =$ "On input x ,
 - If $x \neq w$, reject.
 - If $x = w$, then run M on w and accept if M does, else reject.
 - **Run R on $\langle M_w \rangle$. If R accepts, reject; if R rejects, accept.**
- **Question.** Is this a mapping reduction?

No Mapping Reduction: A_{TM} to E_{TM}

- Earlier reduction is mapping reduction from A_{TM} to $\overline{E_{\text{TM}}}$
 - That is, $\overline{A_{\text{TM}}} \leq_m E_{\text{TM}}$
- What can we say about E_{TM} ?
 - Since $\overline{A_{\text{TM}}}$ is not Turing recognizable, E_{TM} is also not Turing recognizable.
- Found another example of a language that is not recognizable!
- **Exercise.** Show that A_{TM} is not mapping reducible to E_{TM} .

Undecidability Summary

Question. Which of these are decidable?

- Acceptance problems for DFA, CFG, TM
- Emptiness problems for DFA, CFG, TM
- Accepts all strings problem for DFA, CFG, TM
- Equivalence problems for DFA, CFG, TM

Rice's Theorem

Any nontrivial **property of the languages** recognized by Turing machines is undecidable.

- Is the language empty? Is it finite? Is it infinite? Is it regular?
- Does the language contain strings in Σ^*
- Is the language the same as the language of another TM?

We proved many such examples in class.

HW 6 will have more practice with these.

Rice's Theorem

Any nontrivial **property of the languages** recognized by Turing machines is undecidable.

- Is the language empty? Is it finite? Is it infinite? Is it regular?
- Does the language contain strings in Σ^*
- Is the language the same as the language of another TM?

In contrast, questions about the TM's structure are decidable

- Has more than 15 states, has no transitions into its reject state, etc

Decidable or Not

Questions about behavior of TM's computation on inputs may or may not be decidable.

(HW 8 Problem): One of these is decidable, one is not:

- Does a given TM M and input w , does it ever move its head left when running on w ?
- Does a given TM M and input w , does it ever move its head three times in a row when running on w ?